

Amortized Bayesian parameter recovery for dynamical systems with conditional flow matching

The `amortix` package: architecture, evaluation methodology, and benchmarks

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Abstract

We describe `amortix`, a package for amortized Bayesian estimation of the parameters of dynamical systems — ordinary and stochastic differential equations and discretized PDEs — from noisy observations at a finite set of time points. A conditional flow-matching model is trained once on simulated (parameter, data) pairs and thereafter produces posterior samples for any new dataset without likelihood evaluations or per-dataset optimization; a single trained network handles any number and placement of observation points, the setting required for sequential experiment refinement and Bayesian experimental design. The package couples three components that we argue are only useful together: (i) a set-transformer encoder whose input representation is learnable rather than prescribed — a monotone mixture-of-scales reparametrization of the value axis, attention phases computed from physical observation times, and explicit set-cardinality features; (ii) a benchmark suite of fifteen systems spanning finance, neuroscience, epidemiology, ecology, pharmacokinetics, classical mechanics, and reaction–diffusion PDEs; and (iii) an evaluation harness in which simulation-based calibration is used only as a screen, and the instrument of record is comparison against exact or validated Markov chain Monte Carlo reference posteriors on the same observed points. On the fixed-design suite, 31 of 32 parameters across nine systems pass simulation-based calibration with a mean coverage error of 1.3 percentage points; on the variable-design suite, posterior widths match exact references to within 5–7% uniformly over design sizes $K = 5\text{--}100$, and the two residual defects found anywhere in the suite decay as measured power laws of the training budget. We compare against neural posterior estimation and flow-matching posterior estimation from the `sbi` package, both network families of BayesFlow 2, and a port of the Simformer, on three systems with exact or validated references. On a multiplicative process, all external configurations either inflate the volatility posterior by factors of 1.4–3.2 or become overconfident (widths 0.4–0.9) depending on the input coordinates they are given, while the learnable representation recovers the exact posterior from raw data. On an additive control process the same external tools improve markedly — the systematic bias disappears and the width inflation shrinks from 2.5–3.2 to 1.6–1.9 — which localizes the dominant failure to input conditioning while showing that a residual representation gap remains even in the benign case. We document this failure mechanism — the conditioning of the encoder input — in three distinct forms, together with learnable modules that repair each one.

1 Introduction

Estimating the parameters of a differential-equation model from noisy, irregularly sampled trajectory data is among the most common inverse problems in applied science. Two features make the fully Bayesian version of this problem expensive. First, for most stochastic and partially observed systems, the likelihood $p(\mathcal{D} \mid m)$ requires integrating over latent paths between observations and is intractable or costly. Second, modern workflows evaluate posteriors *repeatedly*: calibration studies process thousands of synthetic datasets; sequential experiments update the posterior after every new measurement; Bayesian experimental design compares candidate observation schedules by the posteriors they would produce. Per-dataset MCMC or optimization inside such loops is the dominant cost even when a likelihood is available, and is impossible when it is not.

Simulation-based inference (SBI) addresses both issues by learning a conditional generative model $q_\theta(m \mid \mathcal{D})$ from simulated pairs $(m_j, \mathcal{D}_j)$, amortizing the cost of inference into a one-time training phase (Cranmer et al., 2020). Neural posterior estimation (NPE) and its sequential variants (Papamakarios

and Murray, 2016; Greenberg et al., 2019) fit normalizing flows by maximum likelihood; flow-matching and diffusion variants (Wildberger et al., 2023; Gloeckler et al., 2024) replace the density estimator with transport-based generative models. Mature software exists: the **sbi** toolkit (Boelts et al., 2024), BayesFlow (Radev et al., 2020), and the **sbibm** benchmark collection (Lueckmann et al., 2021).

This report describes **amortix**, a compact package (about 3,500 lines, PyTorch, one small default model of 4.8×10^5 weights) built around three design decisions that distinguish it from the packages above.

1. The input representation is learned, not chosen. Conditional density estimators are routinely fitted to a fixed data vector after global standardization. Section 9 shows, across four different generative engines from three codebases, that this convention fails on multiplicative and heavy-tailed processes in a specific, reproducible way, and that supplying hand-derived coordinates trades one failure mode for others (over-dispersion, overconfidence, or bias, depending on the package). **amortix** instead makes the coordinate system part of the model: a learnable monotone reparametrization of the value axis (Sec. 4.2), observation times entering attention as continuous phases (Sec. 4.3), and explicit set-cardinality features (Sec. 4.4). Each component is motivated by a measured failure and validated by an ablation.

2. One network for all observation designs. **amortix** trains a single model to approximate $p(m \mid S)$ coherently for *every* subset S of measurement points of an experiment — arbitrary sizes and placements, including spatio-temporal sensor designs for PDEs (Sec. 8.2). This is the primitive required for sequential refinement and experimental design, and it changes the training procedure (Sec. 5): observation designs must be resampled fresh at every optimizer step, and the distribution of design sizes becomes a modeling choice with measurable consequences.

3. Calibration screening is separated from verification. Simulation-based calibration (SBC) (Talts et al., 2018) is a necessary-condition test with limited power at practical sample sizes; we measure that limit (Sec. 6) and treat SBC only as a screen. The instrument of record throughout is the comparison of posterior mean and width against an exact or validated-MCMC reference posterior computed on the same observed points. The package ships the reference machinery — closed-form posteriors where they exist, exact-likelihood adaptive Metropolis samplers elsewhere, and chain-validation diagnostics — as part of its public API. The concern that motivates this discipline is documented in the SBI literature: learned posteriors can be confidently wrong in ways that rank-based diagnostics do not detect (Hermans et al., 2022).

The remainder of the report presents the method (Sec. 3), the architecture (Sec. 4), training procedures (Sec. 5), the evaluation methodology (Sec. 6), the benchmark suite (Sec. 7), experimental results including comparisons with **sbi**, BayesFlow, and the Simformer on three systems with exact or validated references (Sec. 8), an analysis of the failure modes encountered during development (Sec. 9), and limitations (Sec. 10). Appendix A defines every benchmark system with its equations, priors, and observation model; Appendix B details the reference posteriors; Appendix C maps the terminology of this report to package options.

2 Problem formulation

Let a dynamical system depend on a parameter vector $m \in \mathbb{R}^d$ with a known prior $p(m)$, uniform on a box in all benchmarks below. The system produces a state trajectory $x(t)$ — through an ODE $\dot{x} = f(x; m)$, an SDE $dx = f(x; m) dt + \sigma(x; m) dW_t$, or a spatially discretized PDE — and the trajectory is observed through a noisy channel at K points:

$$y_i = h(x(t_i)) + \varepsilon_i, \quad \mathcal{D} = \{(t_i, y_i)\}_{i=1}^K. \quad (1)$$

The observation operator h may expose only part of the state: in the benchmarks below, only the price of an asset while its stochastic volatility is latent, only the membrane voltage of a neuron while its gating variables are latent, or only three spatial sensors of a PDE field. The target is the posterior $p(m \mid \mathcal{D}) \propto p(\mathcal{D} \mid m) p(m)$.

Amortization. Sampling from the joint distribution is easy — draw $m_j \sim p(m)$, simulate, apply the observation channel — and yields training pairs $(m_j, \mathcal{D}_j)$. A conditional generative model trained on such pairs approximates the posterior for every dataset simultaneously; inference for a new dataset is a forward pass.

Design amortization. We additionally require that a single trained network be correct for any number K and placement of observation times (and sensor positions). Formally, for every subset S of the measurement points of one experiment, the network must approximate $p(m \mid S)$; in particular the posterior must widen consistently as points are removed. This object is what sequential refinement and Bayesian experimental design consume: candidate designs are compared by the posteriors they would produce.

2.1 Challenges

Five properties of this problem class shaped the design of the package; each defeated at least one plausible architecture during development.

1. *Intractable likelihoods.* For SDEs observed at gaps, the likelihood integrates over latent paths; for hidden-state models even the observed-process transition density is unavailable. Classical estimators exist per problem but each is a separate derivation (Appendix A lists the one we implement for every benchmark system).
2. *The width of the posterior must be right.* A point estimate can be checked against the truth; a posterior asserts an uncertainty, and the two failure directions are asymmetric in consequence: an overconfident posterior is silently wrong for downstream decisions. Verifying width requires reference posteriors or calibration tests, and the standard calibration test has limited power (Sec. 6), so detecting failure is itself nontrivial (Hermans et al., 2022).
3. *One network must serve all designs.* Posterior width is governed by the number of observations K , yet softmax attention computes normalized averages that are nearly invariant to K (Sec. 4.4); and the training distribution over designs acquires measurable consequences (Sec. 5).
4. *Conditioning of the raw data.* Multiplicative processes have a per-dataset floating scale; jump processes produce squared-increment features whose sample standard deviation exceeds the median scale by six orders of magnitude; the coordinate in which the process is well-behaved (often the logarithm) differs per problem. Section 9 traces every hard failure encountered in this project to this property.
5. *Weak identifiability.* Posteriors of real systems concentrate on ridges. In the Fisher–KPP benchmark only the front speed $2\sqrt{Dr}$ is sharply determined, and the trained model’s residual error concentrates along the flat direction of the posterior, where the training signal is weakest (Sec. 8.3).

3 Posterior sampling by conditional flow matching

Parameter normalization. Since the prior factorizes with known one-dimensional CDFs, parameters are mapped componentwise through

$$z = \Phi^{-1}(F_{\text{prior}}(m)) \in \mathbb{R}^d, \quad (2)$$

where Φ is the standard normal CDF. In z -coordinates the prior is exactly $\mathcal{N}(0, I)$ and box constraints disappear. The choice fixes the uninformative limit: when the data carry no information the posterior coincides with the base distribution and the transport below reduces to the identity, so the network need not learn anything to represent ignorance correctly.

Conditional flow matching. The posterior is represented as the endpoint of a learned transport (Lipman et al., 2023). Let $z_0 \sim \mathcal{N}(0, I)$ and let z_1 be the normalized parameter drawn jointly with its dataset $\mathcal{D}$. Define the linear interpolation $z_t = (1 - t)z_0 + tz_1$ and train a velocity network by regression:

$$\mathcal{L}(\theta) = \mathbb{E}_{m, \mathcal{D}, z_0 \sim \mathcal{N}(0, I), t \sim U[0, 1]} \|v_\theta(z_t, t, \mathcal{D}) - (z_1 - z_0)\|^2. \quad (3)$$

The minimizer of (3) is the marginal velocity field whose flow transports $\mathcal{N}(0, I)$ at $t = 0$ to $p(z_1 \mid \mathcal{D})$ at $t = 1$ (Lipman et al., 2023). Sampling integrates $\dot{z} = v_\theta(z, t, \mathcal{D})$ with 20 midpoint steps (a finer RK4/60 solver changes no benchmark result; Sec. 8.3) and maps back through the inverse of (2). Evaluation uses an exponential moving average of the weights. The same estimator family, with a different conditioning architecture, is available in `sbi` as FMPE (Wildberger et al., 2023); the comparison of Sec. 8.4 therefore isolates the effect of the input representation from the choice of generative engine.

No likelihood evaluations, simulator gradients, or per-dataset optimization are required at any stage; the simulator is executed only forward, during training-set generation.

4 Architecture

The network consists of a set encoder that maps the observation set $\mathcal{D}$ to context vectors and a velocity network that consumes them. The default model has width 64, three encoder blocks, three velocity blocks, and 4.8×10^5 parameters in total; Sec. 8.3 shows that capacity is not the accuracy bottleneck on any benchmark, so the default is deliberately small. Throughout, the design constraint is that no transformation of the raw data be problem-specific.

4.1 Tokenization: point tokens and pair tokens

Each observation contributes one token. Two constructions are used.

Point tokens (general case). Observation i becomes $[t_i, w(y_i), \text{sensor id}, \text{set-size feature}]$, where w is the learnable reparametrization of Sec. 4.2. Nothing is assumed about the process.

Pair tokens (Markov observations). When the *observed* coordinate is itself a Markov process — scalar SDEs observed directly — the likelihood factorizes over consecutive observations, and the factorization dictates a sufficient token content.

Proposition 1. *Let the observed process be Markov and let $t_{(1)} < \dots < t_{(K)}$ be the sorted observation times. Then $p(\mathcal{D} \mid m) = p(y_{(1)} \mid m) \prod_{i=1}^{K-1} p(y_{(i+1)} \mid y_{(i)}; m)$, and each factor depends only on the pair $(y_{(i)}, y_{(i+1)})$ and the gap $\Delta t_{(i)}$.*

Pair tokens present exactly these arguments in learnable coordinates,

$$[t_{(i)}, w(y_{(i)}), \Delta w_i, (\Delta w_i)^2, w_t(\Delta t_{(i)})], \quad \Delta w_i = w(y_{(i+1)}) - w(y_{(i)}),$$

where the squared increment supplies the quadratic-variation statistic that diffusion coefficients require. Two additions make the construction robust on heavy-tailed processes (Sec. 9): squared increments are logarithmically compressed before feature normalization, and a set-conditioning block — statistics pooled over the token set modulating each feature channel, initialized to the identity — adapts the normalization to each dataset.

Class rule. Pair tokens are used exactly when the observed process is Markov; point tokens otherwise. The rule was tested in both directions. A pair-token model transfers across Markov systems (a model class trained on geometric Brownian motion re-trains cleanly on Ornstein–Uhlenbeck with unchanged settings); on the Heston model, where the observed price is not Markov because volatility is latent, pair and point models are statistically indistinguishable across all 25 evaluation cells, as the factorization argument predicts — the pair stencil confers an advantage only where the proposition applies. The package selects the construction from a per-problem Markov flag.

4.2 Learnable monotone reparametrization

All tokenizations share one module: a monotone map applied to observed values and to time gaps,

$$w(x) = \sum_k a_k \operatorname{sign}(x) \log(1 + |x|/s_k), \quad a_k \geq 0, \quad (4)$$

with learnable weights a_k over a geometric bank of scales s_k (eight octaves by default). The mixture interpolates continuously between the identity (s_k large) and the logarithm (s_k small), so the network can learn the coordinate in which the process is well-conditioned — for example the log-price coordinate of geometric Brownian motion — rather than having it prescribed.

The specific parametrization matters more than its expressive power, which is the reason we document it. We first attempted a Yeo–Johnson layer (Yeo and Johnson, 2000) with learnable λ : its parameter space contains the logarithm exactly (at $\lambda = 0$), yet across repeated runs on GBM, gradient descent left λ near its initialization and the volatility bias persisted at +0.48 posterior standard deviations — the same failure as with no transform at all. The mixture (4), whose components are all active at initialization, reached the log coordinate reliably. Membership of the correct transform in the hypothesis class is not sufficient; it must be reachable by gradient descent from the initialization, and (4) is built so that it is.

4.3 Observation times as attention phases

Transformers encode order through positional embeddings; rotary embeddings (Su et al., 2024) encode the token *index* as a rotation of query–key pairs. For irregularly sampled data the index is not meaningful, and using it is actively harmful in a way that is easy to miss. In an early configuration, training subsampled long token sequences in place (leaving kept tokens at their original, spread-out indices) while inference presented short, contiguously packed sequences. No individual component was incorrect, yet the mismatch between the two positional distributions produced systematic biases of 1–3 posterior standard deviations at evaluation (Table 3). The failure is a hidden train/test contract on token placement.

amortix removes the contract by computing the attention rotation phases from *physical time*: the rotation angle of head-dimension j for a token observed at time t is $\omega_j t$ with a geometric frequency bank ω_j , so attention logits depend on time differences $t_i - t_j$ and not on storage order. The encoder is then permutation-invariant by construction; we verify 10^{-8} -level invariance under random token permutations at float precision.

4.4 Set size under normalized attention

Posterior width is governed by the number of observations ($\propto K^{-1/2}$ in regular models), but softmax attention computes weighted *means* over the set, and means are nearly invariant to K . The deficit is measurable. We fit linear probes from the encoder’s pooled representation to known sufficient statistics of the GBM benchmark: every architectural variant recovers the posterior-mean statistics ($R^2 = 0.83\text{--}0.94$), but $R^2(\log K)$ separates the variants that produce correct widths from those that do not — 0.985 for pair tokens, 0.957 for index-based positions with contiguous packing, 0.712 for a continuous time-kernel attention bias, 0.498 for pairwise edge-message attention — and the ranking of width errors follows the ranking of $R^2(\log K)$ exactly (Table 3).

The repair requires no process knowledge, only making the set size available: either an explicit $\log K$ input feature, or two learned register tokens (Darcet et al., 2024) appended to every set, which the model uses to accumulate a count. On the hardest cell (dense designs, $K = 100$), the explicit feature reduces the width ratio from 1.95 to 1.49 with bias unchanged at zero. The residual after the count repair is a budget effect, not an architectural one (Sec. 8.3). This analysis also explains an otherwise puzzling observation: index-based positions with contiguous packing outperform metrically correct continuous-time variants at dense designs, because with contiguous packing the maximal position *is* K — the ordinal embedding leaks the count.

4.5 Velocity network

The flow state z_t is processed by self-attention over the d parameter slots. The flow time t and the pooled data context enter every block through adaptive layer norms with zero-initialized modulation (adaLN-Zero, Peebles and Xie 2023), so training starts from an identity-like map and departs only as driven by data. Parameter-slot attention lets the model express posterior correlations (the ridge of the Fisher–KPP posterior, the drift–volatility coupling of GBM) without a joint output head.

5 Training procedures

Common settings. Adam with cosine decay and linear warmup, base learning rate 3×10^{-4} , batch 64, gradient clipping, EMA decay 0.999. Each optimizer step draws fresh flow times t (stratified) and fresh base noise. Fixed-design models train on n simulated datasets for a fixed number of epochs (the gallery of Sec. 8.1 uses $n = 40,000$ and 35 epochs).

Design amortization. Training data are simulated once on a fine grid; observation designs are then resampled *from the stored fine trajectories at every optimizer step*. The procedure has three components, each fixed by a controlled comparison on three systems (GBM, Ornstein–Uhlenbeck, FitzHugh–Nagumo):

1. *Fresh designs at every step.* Re-drawing each trajectory’s observation subset at each step makes design memorization impossible and lets one simulation serve arbitrarily many designs. The alternative — one fixed design per trajectory — closes the dense-design error at long budgets but overfits the design set, producing overconfidence at sparse K (width ratio 0.86 against the exact reference).

2. *The design-size law.* K is drawn 50% log-uniform on $[K_{\min}, K_{\max}]$ and 50% uniform on the dense half of the range. Log-uniform sampling alone places $\approx 6\%$ of the mass at $K \geq 100$ and starves the dense regime, which appears as inflated widths precisely there; the mixture alone halves the dense-design excess (GBM $1.27 \rightarrow 1.14$, OU $1.34 \rightarrow 1.14$; the FitzHugh–Nagumo dense- K SBC failure at $p < 10^{-3}$ disappears, $p = 0.54$).
3. *Budget scaling.* When higher accuracy is required, simulations and optimizer steps are scaled together at constant epochs; Sec. 8.3 shows the residuals then follow power laws in this single budget variable.

With this procedure, the GBM posterior width matches the exact per-design posterior to 5–7% uniformly over $K = 5, 9, 20, 100$ (Fig. 1).

6 Evaluation methodology

Two instruments with different roles are used throughout, plus a diagnostic probe.

6.1 Simulation-based calibration as a screen

SBC (Talts et al., 2018) tests a necessary condition: for $m^* \sim p(m)$ and $\mathcal{D} \sim p(\mathcal{D} | m^*)$, the rank of each component of m^* among posterior samples must be uniform. We run SBC per parameter (uniformity test on 500 simulated datasets with 200 posterior draws each unless stated; $p > 0.05$ passes) and report coverage of central 50% and 90% intervals.

The power of the test at this size is limited, and because our conclusions depend on knowing that limit, we measured it: on cells where an exact reference is available, SBC at 300–500 datasets fails to reject posterior width ratios up to ≈ 1.3 , and its per-cell p -values are unstable between retrains when residual biases are in the range 0.1–0.2 posterior standard deviations. SBC also cannot penalize a posterior that ignores the data and returns the prior; the prior-mean and ridge controls of Sec. 7 exist for that reason. SBC is therefore used as a cheap screen, never as a verdict.

6.2 Reference posteriors as the record

Wherever a tractable likelihood exists, we compare distributions on the same observed points using two statistics per parameter: the signed error of the posterior mean in units of the reference posterior standard deviation (*bias*) and the ratio of standard deviations (*width*; 1 is exact, > 1 conservative, < 1 overconfident). References are of three kinds (full details in Appendix B):

- *Closed form.* The linear-Gaussian benchmark (exact Gaussian posterior) and geometric Brownian motion, whose log-returns give a conjugate normal–inverse- χ^2 posterior importance-corrected to the box prior.
- *Exact-likelihood MCMC.* An adaptive random-walk Metropolis sampler (after Haario et al., 2001): proposal scale adapted to a 0.25 acceptance target during burn-in, proposal shape replaced by the empirical covariance Cholesky factor rescaled to $2.38/\sqrt{d}$, adaptation frozen before the retained phase, out-of-box proposals rejected (which samples the box-truncated target exactly). Likelihoods are exact per-gap transition densities: Gaussian Euler–Maruyama chain transitions for OU (matching the simulator that generated the data, plus the stationary density of the informative initial state), a Poisson-mixture transition density for the Merton jump-diffusion, log-normal residuals around the Bateman curve for pharmacokinetics, and a Gaussian observation model around the deterministic PDE solve for Fisher–KPP — in the last case one likelihood evaluation is one PDE solve.
- *Chain validation.* A reference is used only after validation. Every MCMC reference in this report is run at least twice from well-separated starting points, and the between-chain discrepancy of posterior means (in posterior-sd units) is reported with the result it certifies. Where a closed form and a chain both exist, the chains reproduce it to a worst mean discrepancy of 0.099 sd and width ratios 0.91–1.03 over 24 comparisons.

Twice during development the screen and the record contradicted each other; both times the contradiction exposed an implementation error (a padding mask leaking into set-level statistics; the positional contract of Sec. 4.3). We consider the redundancy a feature of the methodology rather than an inefficiency.

system	type	d	obs.	design	reference posterior
linear-Gaussian	linear map	4	full	fixed	exact Gaussian
Ornstein–Uhlenbeck	SDE	2	full	fixed	exact-likelihood MCMC
geometric Brownian motion	SDE	2	full	fixed	conjugate closed form
Cox–Ingersoll–Ross	SDE	3	full	fixed	— (Euler pseudo-MLE)
double well	SDE	3	full	fixed	—
stochastic Lotka–Volterra	2-d SDE	4	full	fixed	—
FitzHugh–Nagumo	2-d SDE	4	v only	fixed	—
SEIR	4-d SDE	5	I, D	fixed	—
SINDy-SDE (cubic drift)	SDE	5	full	fixed	—
Heston	2-d SDE	5	price only	variable	—
Merton jump-diffusion	jump SDE	5	full	variable	Poisson-mixture MCMC
Hénon–Heiles	Hamiltonian ODE	3	q_1 only	variable	—
Hodgkin–Huxley	4-d ODE	4	V only	variable	—
pharmacokinetics (Bateman)	ODE (closed form)	3	concentration	variable	log-normal MCMC
Fisher–KPP	PDE (64-point grid)	2	3 sensors	variable	PDE-likelihood MCMC

Table 1: The benchmark suite. “obs.” lists the observed coordinate when the state is only partially visible; “design” distinguishes the fixed-grid gallery from variable-design systems, where every dataset carries its own set of (time, sensor) observation points. Dashes in the last column indicate systems where no tractable reference exists — these are evaluated by SBC and controls only, which is precisely the regime that motivates carrying exact-reference systems in the same suite.

6.3 Encoder probing

To separate encoder failures from flow failures, we fit ridge regressions from the pooled encoder output to known sufficient statistics of the problem. High probe R^2 with poor posterior accuracy localizes the defect downstream of extraction. In the Merton analysis of Sec. 9, the failing configuration extracted robust volatility statistics *better* than the healthy one ($R^2 = 0.967$ on truncated realized variance), redirecting the repair from the encoder’s capacity to the stability of its input representation.

7 Benchmark suite

The suite contains fifteen systems (Table 1): nine with a fixed observation design used for the calibration gallery, and six with variable designs used for design amortization. Every system ships with a classical estimator (`sota()`) and, where a tractable likelihood exists, a reference posterior. Full equations, priors, and observation models are given in Appendix A.

Controls and classical baselines. Every result is read against: (i) a prior-mean predictor (exactly 25% mean absolute error under a uniform prior) and a degree-2 ridge regression on ~ 20 generic summary statistics — these two detect prior-returning posteriors that SBC passes; (ii) the per-system classical estimator: exact MLE for OU and GBM, Euler pseudo-likelihood for CIR, Kramers–Moyal and SINDy-style regression for drift discovery (Brunton et al., 2016), multi-start nonlinear least squares for FitzHugh–Nagumo, SEIR, and Lotka–Volterra, and the exact posterior mean for the linear-Gaussian case; (iii) the reference posteriors of Sec. 6.2.

8 Experimental results

8.1 Fixed-design gallery

Table 2 summarizes the nine fixed-design systems. One model configuration (the default of Sec. 4) is trained per system on 40,000 simulations for 35 epochs; SBC uses 500 datasets with 200 posterior draws. Of 32 parameters, 31 pass; the single failure is the β parameter of stochastic Lotka–Volterra at $p = 0.04$, marginal at the threshold. Mean central-interval coverage error is 1.3 percentage points. On the systems with exact references, the learned posteriors carry 89–90% of the available information (width relative to the exact posterior). Per-parameter p -values are listed in Appendix A.

system	d	coverage err.	SBC pass	cov. 50%	cov. 90%
linear-Gaussian	4	0.9 pp	4/4	51%	91%
Ornstein–Uhlenbeck	2	1.0 pp	2/2	52%	90%
SEIR	5	1.6 pp	5/5	49%	89%
geometric Brownian motion	2	1.2 pp	2/2	50%	90%
Cox–Ingersoll–Ross	3	1.0 pp	3/3	51%	90%
double well	3	1.8 pp	3/3	49%	89%
stochastic Lotka–Volterra	4	1.7 pp	3/4	51%	89%
FitzHugh–Nagumo	4	1.4 pp	4/4	50%	90%
SINDy-SDE (cubic drift)	5	1.2 pp	5/5	49%	90%
total / mean	32	1.3 pp	31/32	50%	90%

Table 2: Fixed-design gallery: one default model per system, 40,000 simulations, 35 epochs. Coverage error is the mean absolute deviation of central-interval coverage from its nominal level; “SBC pass” counts parameters with uniformity $p > 0.05$ (500 datasets $\times$ 200 draws).

encoder variant	σ width ratio at K				$R^2(\log K)$
	5	9	20	100	
point tokens, index positions, spread subsampling	0.94*	1.13*	1.42*	2.08*	—
point tokens, index positions, contiguous	1.05	1.17	1.13	1.73	0.957
point tokens, time-kernel attention bias	1.03	1.14	1.31	2.51	0.712
point tokens, edge-message attention	1.03	1.15	1.29	2.45	0.498
point tokens, time phases	1.01	1.10	1.14	1.95	—
+ explicit $\log K$ feature	1.01	1.05	1.08	1.49	—
pair tokens, time phases	1.02	1.02	1.02	1.27	0.985
pair tokens + full training procedure	1.06	1.06	1.05	1.07	—

Table 3: GBM with variable designs: posterior width for σ against the exact per-design posterior (96 datasets per bucket) and, where measured, the probe R^2 for $\log K$ from the pooled encoder output. *This row also carries biases of +1.09 to +3.30 posterior sd (the positional train/test mismatch of Sec. 4.3); all other rows have biases consistent with zero.

8.2 Variable designs

Architecture comparison under a common protocol. Table 3 and Fig. 1 compare encoder variants for design amortization on GBM, trained at $K \sim \log\text{-uniform}[2, 128]$ random-time designs and evaluated against the exact per-design posterior (96 datasets per bucket). Three observations organize the table. First, index-based positions combined with in-place subsampling during training (leaving kept tokens at spread positions while inference packs contiguously) produce biases of +1.1 to +3.3 posterior sd with no individually incorrect component — the positional pathology of Sec. 4.3; repacking or replacing positions with time phases removes the bias entirely. Second, all point-token variants share inflated widths at dense designs, in proportion to how poorly their pooled representation encodes $\log K$ (rightmost column; the probe of Sec. 4.4) — including two attention mechanisms specifically constructed to exploit continuous time (a learned per-head time-kernel logit bias in the spirit of relative-position methods, and edge-message attention carrying gap-modulated increments on all pairs). Third, pair tokens with time-phase attention, plus the training procedure of Sec. 5, bring every bucket to within 7% of the exact width.

Variable-design zoo. With the class rule and training procedure fixed, six further systems were trained without per-system adjustments. Heston (five parameters, latent volatility): all 25 SBC cells pass, and pair tokens give no advantage over point tokens, consistent with the class boundary (Sec. 4.1). Merton jump-diffusion: 25 of 25 cells after the heavy-tail treatment of Sec. 9, including dense-design cells that failed at $p < 10^{-3}$ before it; the residual width sits on the budget curve below. Hénon–Heiles (parameters of the potential of Lubich et al. 2015, observing one coordinate at random times): calibrated at every design size down to $K = 6$, where the posterior widens honestly rather than degrading. Hodgkin–Huxley (Hodgkin and Huxley, 1952): calibrated on mixed designs; one dense-design cell shows the budget signature. Pharmacokinetics: at $K = 6$ irregular blood draws, bias -0.01 to -0.06 sd and widths 1.09–1.33 against exact-likelihood MCMC. Fisher–KPP: the posterior concentrates on the ridge $D \cdot r \approx \text{const}$; the residual along-ridge bias decays with budget (next subsection).

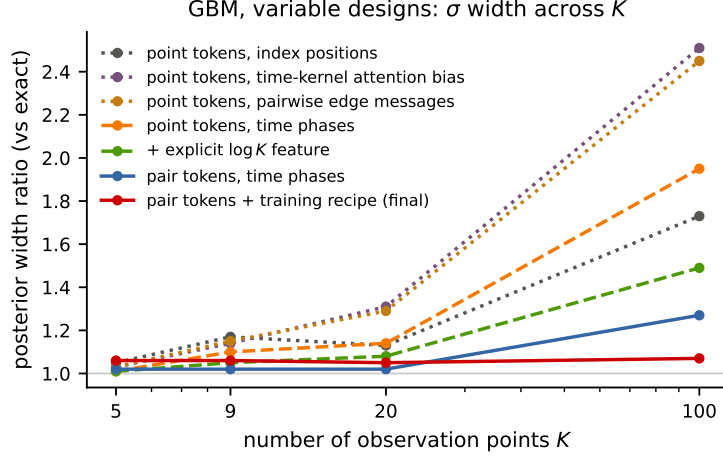


Figure 1: Posterior width for the GBM volatility against the exact per-design reference, as a function of design size K , for the encoder variants of Table 3. Dotted: point-token variants without set-size information; dashed: partial repairs; solid: pair tokens, and pair tokens with the full training procedure of Sec. 5.

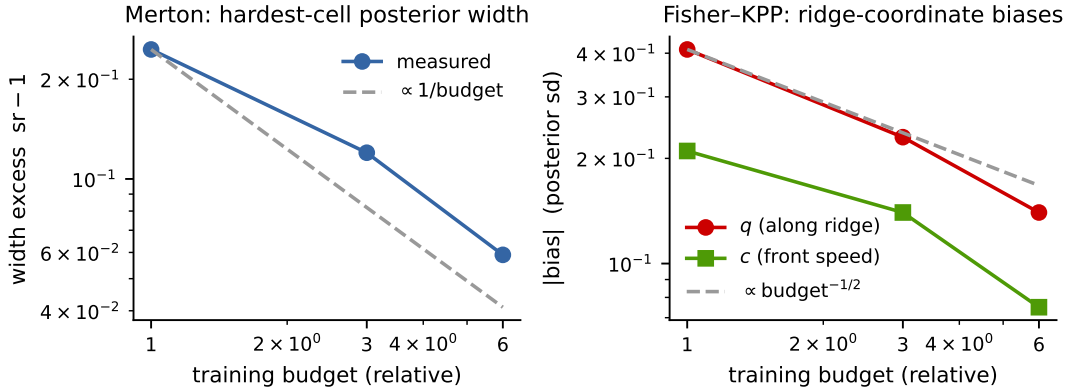


Figure 2: The two residual defects of the suite against training budget (simulations and steps scaled together; model size fixed). Left: Merton width excess on the hardest evaluation cell, with the $1/\text{budget}$ guide. Right: Fisher-KPP posterior-mean biases in the ridge coordinate system, with the $\text{budget}^{-1/2}$ guide.

8.3 Residuals scale as power laws in the training budget

Two residual defects remained anywhere in the suite after the architectural work: inflated width on the hardest Merton cell, and along-ridge bias on Fisher-KPP. Both were driven down by scaling simulations and optimizer steps together at fixed model size (Fig. 2). The Merton width excess follows $\propto 1/\text{budget}$ ($0.246 \rightarrow 0.120 \rightarrow 0.059$ at $\times 1, \times 3, \times 6$); the KPP ridge bias follows $\approx \text{budget}^{-1/2}$ ($+0.41 \rightarrow +0.23 \rightarrow +0.14$ sd along the ridge, with the sharply identified front-speed component decaying in parallel). Two negative controls exclude the obvious alternatives: doubling network width and depth at fixed budget changes the Merton cell by less than 0.02 in width ratio (1.103 vs 1.120 at $\times 3$), and refining the sampling ODE solver from 20 midpoint steps to 60 RK4 steps changes nothing measurable. The weakly identified posterior direction carries the weakest regression signal in (3) and converges last; we find no evidence of a floor within the measured range.

8.4 Comparison with existing implementations

Protocol. We compare against the directly comparable amortized estimators available in maintained packages: NPE with its default masked autoregressive flow (Papamakarios et al., 2017) and FMPE (Wildberger et al., 2023) from `sbi` 0.27 (Boelts et al., 2024); the coupling-flow and flow-matching networks of BayesFlow 2.0 (Radev et al., 2020); and, because it is the closest architecture to ours in the literature, a port of the Simformer (Gloeckler et al., 2024) built from its authors’ minimal example (transformer-diffusion over the joint, random condition masks, their architecture and SDE settings, full 75,000-step

method (input)	train	inference	μ : bias / width	σ : bias / width
amortix CFM, raw prices	488 s	426 ms	+0.00 / 1.00	−0.00 / 1.05
sbi NPE, raw prices	29 s	13 ms	−0.02 / 1.06	+0.45 ± 0.19 / 2.64
sbi NPE, log-prices	34 s	12 ms	−0.04 / 1.01	+0.32 ± 0.17 / 2.29
sbi NPE, log-returns	10 s	13 ms	+0.07 / 1.07	+0.13 ± 0.10 / 1.44
sbi FMPE, raw prices	27 s	370 ms	−0.03 / 1.03	+0.20 ± 0.16 / 2.46
sbi FMPE, log-returns	17 s	333 ms	+0.01 / 1.03	−0.02 ± 0.07 / 1.09
BayesFlow 2 coupling, raw prices	384 s	4 ms	−0.04 / 0.95	−0.13 ± 0.11 / 1.85
BayesFlow 2 coupling, log-returns	459 s	6 ms	−0.01 / 0.42	−0.15 ± 0.05 / 0.54
BayesFlow 2 flow matching, raw prices	220 s	2220 ms	+0.06 / 1.00	−0.36 ± 0.11 / 1.99
BayesFlow 2 flow matching, log-returns	296 s	2311 ms	−0.04 / 0.46	−0.30 ± 0.06 / 0.85
Simformer port, raw prices	591 s	1057 ms	−0.04 / 1.08	−0.30 ± 0.20 / 3.16
Simformer port, log-returns	548 s	~10 ³ ms	−0.08 / 1.10	−0.48 ± 0.02 / 1.36
exact-likelihood MCMC	—	42 ms	reference (chain check: −0.04 / +0.01)	

Table 4: GBM, fixed 95-point design, all learners at a 20,000-simulation budget with package defaults, scored on 200 test datasets against the exact conjugate posterior. Bias in units of the exact posterior sd (with standard error where it is distinguishable from zero); width is the sd ratio. Inference time is per dataset for 2,000 posterior samples. Timings on one laptop CPU except the Simformer rows (H200 GPU).

training). All arms receive the same simulation budget (20,000 draws from the same prior and simulator), run with their package defaults and their own stopping rules, and are scored on the same test datasets against the same reference posteriors, using the bias and width statistics of Sec. 6.2. Because these estimators consume fixed-length vectors, the comparison uses fixed observation designs. Three systems are used, chosen to separate the possible explanations of any performance gap: a multiplicative process where input conditioning is hostile (GBM, exact conjugate reference), an additive process where it is comparatively benign (OU, exact-likelihood MCMC reference, two validated chains per dataset), and a PDE where the reference itself is expensive (Fisher–KPP, one PDE solve per likelihood evaluation). Where informative, external arms are additionally given hand-derived inputs — log-prices, or per-gap log-returns, the exact coordinates of the GBM likelihood factorization — to separate the effect of the input representation from the generative engine.

Multiplicative process (GBM). Table 4 and Figs. 3, 4 give the results; timing columns are wall-clock on one laptop CPU (Apple M-series; Simformer rows on an H200 GPU, so their timings are not comparable with the rest while their accuracy rows are). Three regularities are visible. First, with raw prices as input, every external configuration inflates the volatility posterior — widths 1.85 (BayesFlow coupling) to 3.16 (Simformer port) — irrespective of the generative engine; **sbi** itself emits a warning about precision loss during z-scoring on this data, which is a fair description of the mechanism (Sec. 9). Second, supplying hand-derived coordinates does not produce convergent behavior: with identical log-return inputs, NPE remains 44% over-dispersed, FMPE is nearly exact (1.09), both BayesFlow arms become overconfident (widths 0.42–0.85), and the Simformer port acquires a −0.48 sd bias. Each of these defects is invisible without an exact reference, and each lies within or near the blind region of SBC at practical sizes (Sec. 6). Third, at fixed input coordinates the flow-matching engine outperforms the autoregressive flow (1.09 vs 1.44), consistent with the engine choice of Sec. 3, but no external arm matches the learnable representation operating on raw data (1.05). The verdicts are stable across devices and seeds: re-running the **sbi** and BayesFlow arms on an H200 GPU with fresh seeds reproduces every failure direction (NPE 2.48, FMPE 2.30, BayesFlow coupling 1.80 on raw input; BayesFlow overconfident at 0.40–0.72 on log-returns).

Additive control (Ornstein–Uhlenbeck). The same protocol on OU — an additive process with no floating scale — separates the candidate explanations of the GBM results (Table 5). The reference is exact-likelihood MCMC (per-gap Euler–Maruyama transitions plus the stationary initial density; 62 ms per chain; two-chain validation: worst mean discrepancy 0.29 sd, mean 0.08 sd over 100 datasets). The external estimators improve markedly relative to GBM: the volatility bias is no longer significant (+0.19 ± 0.19 for NPE, +0.08 ± 0.12 for FMPE), and the width inflation drops from 2.5–3.2 to 1.9 (NPE) and 1.6 (FMPE). This is the pattern the conditioning analysis predicts: the form-1 pathology of Sec. 9 is absent by construction, and what remains is the general difficulty of assembling quadratic-variation statistics from a raw high-dimensional vector — present but smaller. **amortix** reads 1.03/1.05 on the same data.

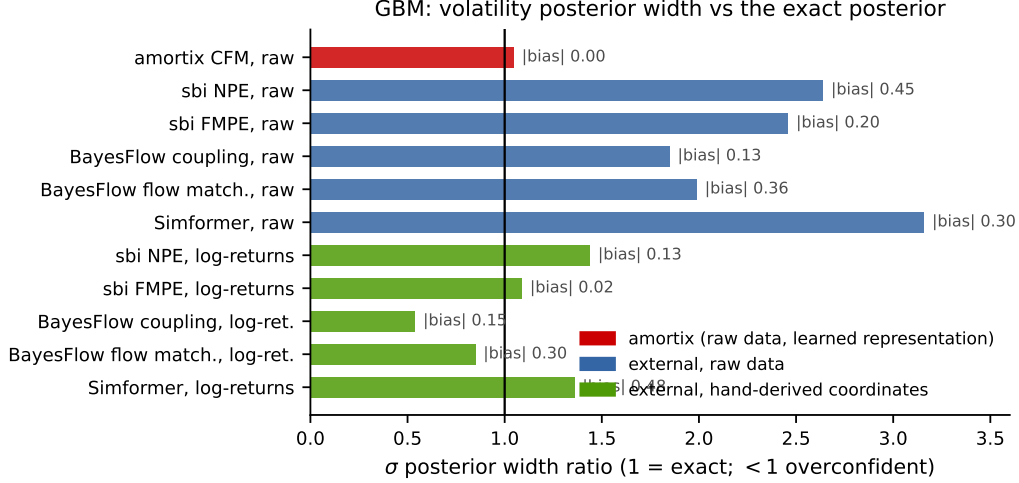


Figure 3: Volatility posterior width (ratio to the exact posterior) for all arms of Table 4. Values below one are overconfident. Annotations give the absolute posterior-mean bias in posterior-sd units.

method (raw values as input)	train	inference	θ : bias / width	σ : bias / width
amortix CFM	183 s	32 ms	−0.01 / 1.03	+0.01 / 1.05
sbi NPE	128 s	10 ms	+0.07 / 1.15	+0.19 ± 0.19 / 1.88
sbi FMPE	87 s	147 ms	+0.11 / 1.01	+0.08 ± 0.12 / 1.60
exact-likelihood MCMC	—	124 ms	reference (two chains per dataset)	

Table 5: Ornstein–Uhlenbeck, fixed 74-point design, 20,000-simulation budget, 100 test datasets, one H200 GPU (MCMC on its CPU; time given for the two validation chains). The additive control: external biases vanish and widths improve but remain inflated; compare Table 4.

Scope of the comparison. Every external arm ran at its package defaults under our simulation budget; specialists could improve each with problem-specific embedding networks or summary statistics. The comparison is designed to measure exactly that dependence: what accuracy is delivered when the input representation is not hand-engineered per problem. The Simformer arm in particular is a port of a minimal example rather than a tuned configuration, and its numbers should be read as representative of that recipe, not of the method’s ceiling. Conversely, the variable-design capability of Sec. 8.2 is not exercised here at all, since the external estimators consume fixed-length inputs by default.

8.5 Computational cost

The default model trains in minutes to tens of minutes per system: on one H200 GPU, a variable-design zoo system at base budget (20,000 simulations, 12,000 steps) takes 8–12 minutes, and the $\times 6$ budget behind the sharpest numbers of Fig. 2 takes about 50 minutes; on a laptop CPU the fixed-design gallery systems take 8–45 minutes. Inference draws 2,000 posterior samples for a dataset in 426 ms on the laptop CPU and 72 ms (batched; 135 ms single) on the GPU. GPU and CPU runs produce bit-identical posteriors because all stochastic generation is kept on a CPU random number generator by design, so a device change cannot silently alter results. For orientation against non-amortized inference: on GBM, whose likelihood is a three-line formula, adaptive Metropolis costs 42 ms per dataset and amortization buys nothing for a single dataset; on Fisher–KPP, where one likelihood evaluation is one PDE solve (16 ms), the same reference sampler costs minutes per dataset (measured in Table ??); and for Heston or Hodgkin–Huxley no likelihood formula is available at any price. The regime where amortization pays is determined by the likelihood cost, the number of datasets, and — as Table 4 indicates — by whether a trustworthy input representation for a conditional density estimator is available without one.

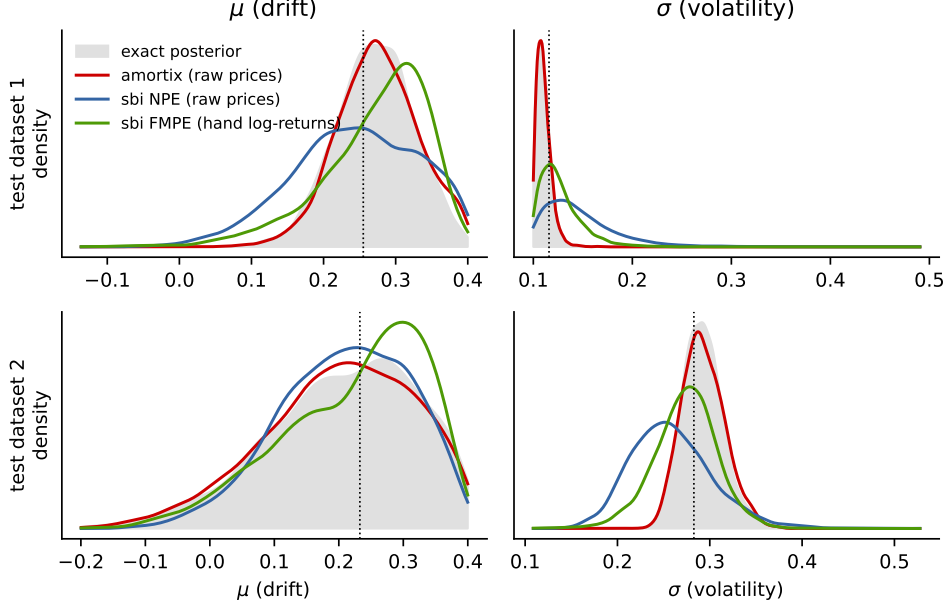


Figure 4: Posterior marginals on two representative GBM test datasets (rows) for both parameters (columns): exact posterior (shaded), `amortix` from raw prices, `sbi NPE` from raw prices, and `sbi FMPE` from hand-computed log-returns. Dotted vertical lines mark the true parameters.

9 Failure analysis: conditioning of the encoder input

Every accuracy failure encountered during the development of this package — including, we argue, the external failures of Table 4 — reduces to a single cause appearing in three forms: the conditioning of the numbers presented to the set encoder. We document the three forms with their measurements and repairs; all three repairs are learnable modules rather than problem-specific transformations.

Form 1: floating multiplicative scale. On GBM, raw price tokens force the network to divide squared increments by squared levels internally; with global standardization this manifests as a volatility bias of $+0.48$ posterior sd (confirmed against the closed-form posterior) that no amount of training removes. The repair is the monotone mixture reparametrization of Sec. 4.2, which learns the log coordinate; the Yeo–Johnson ablation there shows why the parametrization, not the hypothesis class, is the operative choice. External estimators exhibit the same form on the same data (Table 4, raw-price rows).

Form 2: cardinality blindness. Normalized attention erases set size, which controls posterior width under design amortization; the probe measurements and the $\log K$ /register-token repairs are given in Sec. 4.4, with the arm-by-arm consequences in Table 3.

Form 3: heavy tails. On the Merton jump-diffusion, squared increments have sample standard deviation 3.1×10^4 against a diffusion signal of order 10^{-2} –1; running feature normalization never converges, and the flow trains against a drifting representation, producing a $+1.12$ sd volatility bias. Encoder probing (Sec. 6.3) shows the information is extracted — the failing configuration encodes robust volatility statistics better than the healthy one — so the repair targets representation stability: logarithmic compression of increment features (near-identity on clean diffusions since $\log(1+x) \approx x$ for small x) restores all 25 evaluation cells. A precision follow-up against the Poisson-mixture reference then isolated a smaller ($+0.27$ sd) residual whose mechanism is a jump-classification threshold that floats with the unknown volatility; a per-dataset set-conditioning module (pooled statistics modulating feature channels, initialized to the identity) removes it as effectively as a hand-crafted per-dataset rescaling ($+0.127 \pm 0.055$ vs $+0.094 \pm 0.054$ sd, statistically indistinguishable). This was one of three such paired comparisons in the project (log coordinate, set size, robust scaling); in all three the learnable module matched the hand-crafted repair once initialized at the working configuration. We summarize the pattern as a design principle: the hand-crafted fix identifies the mechanism and the initialization; the learnable module placed there is what ships.

Three implementation conditions proved necessary for the set-conditioning module and are recorded here because each was found through a failure: the modulation must be zero-initialized (training must start from the working baseline); it must act after feature normalization, not before (the reverse ordering re-creates the representation drift it is meant to cure); and pooled set statistics must exclude padding positions (a masking error here made padded and packed representations of the same dataset inequivalent, detected only because SBC and the exact-reference probe disagreed).

10 Discussion and limitations

The evidence assembled here supports a specific claim: for dynamical-systems inverse problems, the input representation of an amortized posterior estimator is not a preprocessing detail but the dominant accuracy factor, and it can be made part of the model rather than part of the problem specification. The comparison of Sec. 8.4 shows the cost of the alternative — fixed representations with global standardization — across four generative engines, on a problem where the correct answer is known in closed form. The OU control separates the mechanisms: on additive data the bias of the external estimators disappears and their width inflation drops by roughly a factor of two but does not vanish, so the conditioning pathology accounts for the dominant part of the failure and a smaller representation-learning gap accounts for the rest.

Limitations. (i) The remaining inaccuracies of the suite decay as measured power laws in training compute (Fig. 2); we have not searched for methods that improve the exponents. (ii) The power limits of SBC imply that small residuals can only be certified where an exact or validated reference exists; systems without tractable likelihoods inherit the weaker screen, and carrying reference-equipped systems alongside them in one suite is our current mitigation. (iii) The comparison of Sec. 8.4 uses package defaults; it measures the out-of-the-box regime, not the ceiling of any method. (iv) Bayesian experimental design — the intended consumer of design amortization — is not yet demonstrated; the training procedure of Sec. 5 was built so that the required object $p(m | S)$ is already coherent across subsets S . (v) No real-data study with posterior predictive checks is included yet. (vi) All priors are uniform on boxes; the probit normalization (2) extends to any factorized prior with tractable CDFs, but correlated priors would require a different normalization.

A Benchmark system definitions

All SDEs are integrated by Euler–Maruyama on the stated fine grid; ODEs by RK4 (Hodgkin–Huxley by a semi-implicit scheme for the gating variables); the PDE by finite differences. Fixed-design observations are token sets over the union of the stated channels; “fast/slow” channels observe every step over a short window and every j -th step over the horizon respectively. Priors are uniform on the stated boxes. Classical estimators (`sota()`) are listed per system.

Linear-Gaussian. $y = Am + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.5^2 I_6)$, with a fixed correlated design matrix $A \in \mathbb{R}^{6 \times 4}$; $m \in [-3, 3]^4$. Exact Gaussian posterior; the classical estimator is the posterior mean itself. This system checks the flow engine in isolation from the encoder.

Ornstein–Uhlenbeck. $dX = -\theta X dt + \sigma dW$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$; $dt = 0.02$, 500 steps; channels: fast (1×50), slow (20×24). Prior $\theta \in [0.3, 3]$, $\sigma \in [0.2, 1.5]$. Classical: exact MLE from per-gap Gaussian transitions.

Geometric Brownian motion. $dS = \mu S dt + \sigma S dW$, $S_0 = 1$; $dt = 0.01$, 500 steps (five “years”); channels: fast (1×60), slow (10×40). Prior $\mu \in [-0.2, 0.4]$, $\sigma \in [0.1, 0.6]$. Classical: exact MLE on log-returns. Reference: conjugate posterior (Appendix B).

Cox–Ingersoll–Ross. $dX = a(b - X)dt + \sigma\sqrt{X}dW$; $dt = 0.02$, 500 steps; channels as OU. Prior $a \in [0.3, 3]$, $b \in [0.3, 1.5]$, $\sigma \in [0.1, 0.5]$. Classical: Euler pseudo-likelihood MLE (Cox et al., 1985).

Double well. $dX = (\theta_1 X - \theta_2 X^3)dt + \sigma dW$; $dt = 0.01$, 1000 steps, noise strong enough for barrier crossings; channels: fast (1×80), slow (25×36). Prior $\theta_{1,2} \in [0.5, 3]$, $\sigma \in [0.4, 1.2]$. Classical: Kramers–Moyal regression.

Stochastic Lotka–Volterra. Prey/predator pair with multiplicative noise, $dx_1 = (\alpha x_1 - \beta x_1 x_2) dt + 0.1 x_1 dW_1$, $dx_2 = (\delta x_1 x_2 - \gamma x_2) dt + 0.1 x_2 dW_2$; both components observed; $dt = 0.01$, 600 steps; channels: fast (1×50), slow (15×40). Prior $\alpha, \gamma \in [0.8, 1.5]$, $\beta, \delta \in [0.4, 1.2]$. Classical: multi-start nonlinear least squares.

FitzHugh–Nagumo. $\dot{v} = v - v^3/3 - w + I + \text{noise}$, $\dot{w} = \epsilon(v + a - bw)$; only v observed at 25 evenly spaced times over horizon 40 ($dt = 0.05$), observation noise 0.05. Prior $a, b \in [0.5, 0.9]$, $\epsilon \in [0.05, 0.3]$, $I \in [0, 0.5]$. Classical: multi-start nonlinear least squares on the deterministic solve.

SEIR with mortality. Five-compartment stochastic epidemic (two-phase contact rate $\beta_1 \rightarrow \beta_2$, incubation α , recovery γ_r , mortality γ_d); infected and deceased compartments observed at 20 days over a 60-day horizon ($dt = 0.1$), observation noise 0.004. Prior in Table 6. Classical: nonlinear least squares on (I, D) .

SINDy-SDE. $dX = (c_0 + c_1 X + c_2 X^2 + c_3 X^3) dt + \sigma dW$ (in the spirit of Brunton et al., 2016); $dt = 0.01$, 1000 steps; channels: fast (1×80), slow (20×45). Prior $c_0, c_2 \in [-0.5, 0.5]$, $c_1 \in [-2, 0]$, $c_3 \in [-1, -0.1]$, $\sigma \in [0.3, 0.8]$. Classical: SINDy-style least squares on Kramers–Moyal estimates.

Heston (variable designs). $dS = \mu S dt + \sqrt{v} S dW_1$, $dv = \kappa(\bar{v} - v) dt + \xi \sqrt{v} dW_2$, $\text{corr}(dW_1, dW_2) = \rho$ (Heston, 1993); price observed, volatility latent; $dt = 0.01$, 500 steps, designs of $K \in [2, 128]$ random times. Prior $\mu \in [-0.1, 0.3]$, $\kappa \in [0.5, 5]$, $\bar{v} \in [0.02, 0.3]$, $\xi \in [0.1, 1]$, $\rho \in [-0.9, 0]$.

Merton jump-diffusion (variable designs). $dS/S = \mu dt + \sigma dW + d(\sum_i (e^{J_i} - 1))$ with Poisson jump rate λ and $J_i \sim \mathcal{N}(\mu_J, s_J^2)$ (Merton, 1976); $dt = 0.01$, 500 steps, $K \in [2, 128]$. Prior $\mu \in [-0.1, 0.3]$, $\sigma \in [0.1, 0.5]$, $\lambda \in [0.2, 3]$, $\mu_J \in [-0.3, 0.3]$, $s_J \in [0.05, 0.4]$. Reference: Poisson-mixture transition density (Appendix B).

Hénon–Heiles (variable designs). Two-dimensional Hamiltonian oscillator with the coupling potential used in Lubich et al. (2015), frequencies (ω_1, ω_2) and coupling λ ; q_1 observed at random times with noise; $dt = 0.05$, 800 steps, $K \in [4, 64]$. Prior $\omega_{1,2} \in [0.7, 1.3]$, $\lambda \in [0.02, 0.22]$.

Hodgkin–Huxley (variable designs). Standard four-dimensional membrane model (Hodgkin and Huxley, 1952); voltage observed at random times; parameters (g_{Na}, g_K, g_L, I) with prior $[60, 180] \times [10, 50] \times [0.05, 0.5] \times [4, 12]$; 3000 integration steps, $K \in [4, 96]$.

Pharmacokinetics (variable designs). One-compartment oral dosing (Bateman function), concentration $c(t) = \frac{D_0 k_a}{V(k_a - k_e)}(e^{-k_e t} - e^{-k_a t})$ with dose $D_0 = 500$; log-normal assay noise (10%); horizon 24 h, $K \in [3, 64]$ irregular draws. Prior $k_a \in [0.5, 4]$, $k_e \in [0.05, 0.4]$, $V \in [20, 100]$. Reference: log-normal likelihood MCMC.

Fisher–KPP (variable designs). $\partial_t \theta = D \partial_{xx} \theta + r \theta(1 - \theta)$ on $[0, 1]$ with reflecting boundaries, 64-point finite-difference grid, Gaussian initial bump; three fixed sensors at $x = 16/63, 32/63, 48/63$ observed at random (time, sensor) pairs with noise 0.02; $dt = 0.01$, 200 steps, $K \in [4, 96]$. Prior $D \in [0.001, 0.01]$, $r \in [2, 10]$. Reference: Gaussian observation model around the deterministic solve; one likelihood evaluation is one PDE solve.

B Reference posteriors

GBM conjugate posterior. With observations at indices $i_0 = 0 < i_1 < \dots$ of the fine grid, log-returns $r_k = \log(S_{i_{k+1}}/S_{i_k})$ over gaps τ_k are independent $\mathcal{N}((\mu - \sigma^2/2)\tau_k, \sigma^2 \tau_k)$. In the coordinates $(b, \sigma^2) = (\mu - \sigma^2/2, \sigma^2)$ the model is a Gaussian location–scale family with sufficient statistics $T = \sum_k \tau_k$, $R_1 = \sum_k r_k$, and $SS_w = \sum_k r_k^2/\tau_k - R_1^2/T$; draws from the standard conjugate posterior ($\sigma^2 = SS_w/\chi_{n-1}^2$, $b | \sigma^2 \sim \mathcal{N}(R_1/T, \sigma^2/T)$) are transformed to (μ, σ) and importance-resampled to the uniform box prior with the appropriate Jacobian weight. The sampler was validated against exact-likelihood MCMC on the same designs: over 24 comparisons, worst posterior-mean discrepancy 0.099 sd, width ratios 0.91–1.03.

system	per-parameter SBC p -values				
linear-Gaussian	m_1 : 0.33	m_2 : 0.17	m_3 : 0.70	m_4 : 0.12	
Ornstein-Uhlenbeck	θ : 0.69	σ : 0.60			
SEIR	β_1 : 0.18	β_2 : 0.36	α : 0.50	γ_r : 0.95	γ_d : 0.07
geometric Brownian motion	μ : 0.41	σ : 0.30			
Cox-Ingersoll-Ross	a : 0.80	b : 0.33	σ : 0.96		
double well	θ_1 : 0.68	θ_2 : 0.58	σ : 0.56		
stochastic Lotka-Volterra	α : 0.14	β : 0.04	δ : 0.30	γ : 0.67	
FitzHugh-Nagumo	a : 0.99	b : 0.80	ϵ : 0.38	I : 0.45	
SINDy-SDE	c_0 : 0.47	c_1 : 0.19	c_2 : 0.37	c_3 : 0.11	σ : 0.56

Table 6: Per-parameter SBC uniformity p -values for the fixed-design gallery (500 datasets $\times$ 200 draws; $p > 0.05$ passes). The single failure is marked in bold.

Adaptive Metropolis. Random-walk Metropolis on the box, proposal $x' = x + sC\xi$, $\xi \sim \mathcal{N}(0, I)$. During burn-in (default $\max(500, n/2)$ iterations) the scalar s is adapted toward a 0.25 acceptance rate in blocks of 50 with a decaying Robbins-Monro gain, and once sufficient burn-in states are collected the shape C is replaced by the Cholesky factor of the empirical covariance rescaled by $2.38/\sqrt{d}$ (Haario et al., 2001); adaptation is frozen for the retained phase. Out-of-box proposals are rejected, which samples the box-truncated target exactly since the proposal is symmetric. Likelihoods: per-gap Euler-Maruyama Gaussian transitions for OU (matching the generative process, plus the stationary density of X_0); the exact GBM log-return density; the Poisson-mixture transition density for Merton (truncated at the numerically negligible jump count); log-normal residuals around the Bateman curve; a Gaussian observation model around the deterministic PDE solve for Fisher-KPP. Every chain used as a reference is paired with a second chain from a well-separated start; between-chain posterior-mean discrepancies are reported alongside the results they certify (Sec. 8.4).

C Correspondence with package options

in this report	in the package
point tokens	<code>embed="wpoint"</code>
pair tokens + set conditioning	<code>embed="wfilm"</code>
warped values/increments (fixed designs)	<code>embed="wbasis" / "wdiff"</code>
attention phases from physical time	<code>rope="time"</code>
fresh designs each optimizer step	<code>fit(retokenize=...) via make_retokenizer()</code>
variable-design problem wrapper	<code>amortix.designs.DesignProblem</code>
benchmark systems	<code>amortix.problems (DESIGN_Z00)</code>
calibration screen / reference machinery	<code>sbc_design / amortix.mcmc + likelihood factories</code>
comparison scripts (Sec. 8.4)	<code>examples/baseline_{npe,ou,kpp,bayesflow,simformer}.py</code>

All defaults named in this report are selected automatically by `FlowPosterior(problem)`; the non-default options exist as reproducible controls for the comparisons above.

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